

# Eduard Gorbunov

Tenure-Track Assistant Professor of Statistics and Data Science, MBZUAI, Abu Dhabi, UAE

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## At a Glance

### METRICS

**Google Scholar** 3,821 citations  
**h-index / i10** 35 / 55  
Since 2021: 3,558 citations, h-index 33,  
i10-index 51. Metrics checked July 2026.

### PUBLICATION RECORD

- 74 website-listed publications
- 50 conference papers
- 15 journal papers
- 7 arXiv preprints
- 2 reference entries
- 65 talks and posters

### COLLABORATION FOOTPRINT

- 109 publication collaborators
- 30+ universities/research institutes
- 3+ companies/industrial labs

### RESEARCH AREAS

- Stochastic optimization
- Distributed and federated learning
- Heavy-tailed noise and clipping
- Variational inequalities
- Adaptive and communication-efficient training

### RECOGNITION

- ICML 2024 oral
- NeurIPS 2024 spotlight
- NeurIPS 2020 spotlight
- Ilya Segalovich Award, 2019
- Outstanding reviewer: NeurIPS, ICML, ICLR

## Professional Experience

- 2025–now** Tenure-Track Assistant Professor, Statistics and Data Science, Mohamed bin Zayed University of Artificial Intelligence.
- 2024–2025** Research Scientist, Machine Learning Department, MBZUAI.
- 2022–2024** Postdoctoral Fellow, Machine Learning Department, MBZUAI; hosted by Samuel Horváth and Martin Takáč.
- 2020–2022** Junior Researcher, MIPT/HSE/Yandex Research; Research Consultant, Mila, Montreal.
- 2017–2020** Research positions and internships with Peter Richtárik’s group, Huawei–MIPT, IITP RAS, RANEP–MIPT, and KAUST Visual Computing Center.

## Education

- 2021** PhD in Computer Science, Moscow Institute of Physics and Technology. Thesis: *Distributed and Stochastic Optimization Methods with Gradient Compression and Local Steps*. Advisors: Alexander Gasnikov and Peter Richtárik.
- 2020** MSc in Applied Mathematics and Physics, Moscow Institute of Physics and Technology.
- 2018** BSc in Applied Mathematics and Physics, Moscow Institute of Physics and Technology.

## Research Profile

I develop stochastic and distributed optimization algorithms for machine learning, with emphasis on provable and practical behavior under heavy-tailed noise, communication constraints, local steps, compression, adaptivity, and federated or decentralized systems. Recent work connects clipping, Adam/AdaGrad-type normalization, LMO-based/Muon-type methods, and high-probability and last-iterate analysis to modern training regimes.

## Selected Recent Publications

**Tags:** \* equal contribution; † shared senior authorship; mentored/supervised author markers: <sup>B</sup> BSc, <sup>M</sup> MSc, <sup>P</sup> PhD, <sup>R</sup> RA/Postdoc, <sup>T</sup> Team Member, <sup>V</sup> Visiting Student.

1. A. Shestakov<sup>M</sup>, M. Takáč, E. Gorbunov. **From Optimization to Generalization under Heavy-Tailed Data: The Role of Gradient Clipping**. ICML 2026.
2. E. Shulgin<sup>V</sup>, M. Awad<sup>T</sup>, P. Richtárik, E. Gorbunov. **General Analysis of LMO-based Optimizers: Beyond Bounded Variance**. ICML 2026.
3. R. Islamov<sup>V</sup>, R. Machacek, A. Lucchi, A. Silveti-Falls, E. Gorbunov<sup>†</sup>, V. Cevher<sup>†</sup>. **On the Role of Batch Size in Stochastic Conditional Gradient Methods**. ICML 2026.
4. S. Chezhegov<sup>P</sup>, D. A. Parletta, A. Paudice, E. Gorbunov. **High-Probability Bounds for the Last Iterate of Clipped SGD**. ICLR 2026.
5. S. Khirirat, A. Sadiev<sup>P</sup>, A. Riabinin, E. Gorbunov, P. Richtárik. **Error Feedback under  $(L_0, L_1)$ -Smoothness: Normalization and Momentum**. NeurIPS 2025.
6. S. V. Khah<sup>M</sup>, S. Chezhegov<sup>P</sup>, S. Farahmand, S. Horváth, E. Gorbunov. **Differentially Private Clipped-SGD: High-Probability Convergence with Arbitrary Clipping Level**. AISTATS 2026.
7. S. Chezhegov<sup>P</sup>, Y. Klyukin, A. Semenov, A. Beznosikov, A. Gasnikov, S. Horváth, M. Takáč, E. Gorbunov. **Clipping Improves Adam-Norm and AdaGrad-Norm when the Noise Is Heavy-Tailed**. ICML 2025.

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## Mentoring & Service

### CURRENT TEAM

- 4 MSc students at MBZUAI
- 4 recent visiting students
- Current/recent team members already appear as coauthors in 2026 ICML/UAI papers

### EXTERNAL MENTORING

- 27 unique externally mentored publication contributors in the full-CV role markers
- Breakdown: 2 BSc, 14 MSc, 8 PhD, 3 Research Assistant/Postdoc
- Publication outcomes at AISTATS, ICML, ICLR, NeurIPS, CPAL, EMNLP, MOTOR, and IFAC

### TEACHING

- ML804 Advanced Topics in Continuous Optimization, PhD, MBZUAI
- MTH702 Optimization, MSc, MBZUAI
- Optimization Methods for Machine Learning, MIPT/MADE
- TA/lecturer contributions to ML712 Distributed and Federated Learning

### EDITORIAL SERVICE

- TMLR Action Editor
- Area Chair: NeurIPS, ICLR, ICML
- Co-organizer: ICOMP
- Reviewer for JMLR, Mathematical Programming, SIMODS, SIOPT, TMLR, NeurIPS, ICML, ICLR, AISTATS, NETYS

## Selected Foundational and High-Visibility Publications

**Tags:** \* equal contribution; † shared senior authorship; mentored/supervised author markers: <sup>B</sup> BSc, <sup>M</sup> MSc, <sup>P</sup> PhD, <sup>R</sup> RA/Postdoc, <sup>T</sup> Team Member, <sup>V</sup> Visiting Student.

1. E. Gorbunov\*, N. Tupitsa\*<sup>R</sup>, S. Choudhury<sup>P</sup>, A. Aliev<sup>M</sup>, P. Richtárik, S. Horváth, M. Takáč. **Methods for Convex  $(L_0, L_1)$ -Smooth Optimization: Clipping, Acceleration, and Adaptivity.** ICLR 2025.
2. E. Gorbunov, A. Sadiev<sup>P</sup>, M. Danilova, S. Horváth, G. Gidel, P. Dvurechensky, A. Gasnikov, P. Richtárik. **High-Probability Convergence for Composite and Distributed Stochastic Minimization and Variational Inequalities with Heavy-Tailed Noise.** ICML 2024 oral.
3. K. Mishchenko, E. Gorbunov, M. Takáč, P. Richtárik. **Distributed Learning with Compressed Gradient Differences.** Optimization Methods and Software 2025.
4. E. Gorbunov, M. Danilova, A. Gasnikov. **Stochastic Optimization with Heavy-Tailed Noise via Accelerated Gradient Clipping.** NeurIPS 2020.
5. E. Gorbunov, S. Horváth, P. Richtárik, G. Gidel. **Variance Reduction is an Antidote to Byzantine Workers: Better Rates, Weaker Assumptions and Communication Compression as a Cherry on the Top.** ICLR 2023.
6. E. Gorbunov, N. Loizou, G. Gidel. **Extragradient Method:  $\mathcal{O}(1/k)$  Last-Iterate Convergence for Monotone Variational Inequalities and Connections with Cocoercivity.** AISTATS 2022.
7. M. Ryabinin\*, E. Gorbunov\*, V. Plokhotnyuk, G. Pekhimenko. **Moshpit SGD: Communication-Efficient Decentralized Training on Heterogeneous Unreliable Devices.** NeurIPS 2021.
8. E. Gorbunov, K. Burlachenko<sup>P</sup>, Z. Li, P. Richtárik. **MARINA: Faster Non-Convex Distributed Learning with Compression.** ICML 2021.
9. E. Gorbunov, D. Kovalev, D. Makarenko<sup>P</sup>, P. Richtárik. **Linearly Converging Error Compensated SGD.** NeurIPS 2020 spotlight.
10. E. Gorbunov, F. Hanzely, P. Richtárik. **A Unified Theory of SGD: Variance Reduction, Sampling, Quantization and Coordinate Descent.** AISTATS 2020.

## Current Team

- **MSc students at MBZUAI:** Bilal Ashfaq (co-supervised with Nils Lukas), Mohamed Ayman Mohamed Mohamed Awad, Saurabh Singh, Viktor Kovalchuk (co-supervised with Martin Takáč); all 2025/09–present.
- **Visiting students:** Igor Ignashin (2026/01–2026/03), Savelii Chezhegov (2026/01–2026/03), Rustem Islamov (2025/12–2026/02), Egor Shulgin (2025/10–2025/11).

## Talks, Visibility, and Collaborations

- 65 talks and posters, including an ICML 2024 oral presentation, invited/keynote talks at NETYS, INSAIT, Federated Learning One-World Seminar, MBZUAI Workshop on Collaborative Learning, MLO EPFL, MTL MLOpt, KAUST Rising Stars in AI Symposium, and Russian Optimization Seminar.
- Research visits or collaborations with Mila, KAUST, SIERRA/INRIA, MIPT, EPFL, University of Toronto, HSE, Yandex Research, and other international groups.